

ECN 594: Demographic Interactions and pyblp

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Plan for today

1. **Recap: The demographic model**
2. Typical data
3. The BLP algorithm
4. Classic example: Nevo (2001)
5. Introduction to pyblp
6. Worked example: Basic logit
7. Worked example: With demographics

Recap: The demographic model

- Last time: the demographic interaction model

$$u_{ijt} = \underbrace{x_{jt}\beta + \alpha p_{jt} + \zeta_{jt}}_{\delta_{jt}} + \underbrace{\sum_{k,\ell} \pi_{k\ell} D_{i\ell} x_{jt,k} + \sum_{\ell} \pi_{\alpha\ell} D_{i\ell} p_{jt}}_{\mu_{ijt}} + \varepsilon_{ijt}$$

- δ_{jt} : mean utility (common to all consumers)
- μ_{ijt} : individual deviation based on demographics D_i
- **Key insight:** Demographics $\Rightarrow$ heterogeneous preferences $\Rightarrow$ realistic substitution
- **Today:** Estimate this model using pyblp

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Estimation: What parameters do we need?

- Linear parameters:
 - β : coefficients on product characteristics
 - α : price coefficient
- Nonlinear parameters:
 - π : coefficients on demographic interactions
- Full parameter vector: $\theta = (\alpha, \beta, \pi)$
- Why “linear” vs “nonlinear”?
 - β, α enter the mean utility δ_{jt} linearly
 - π enters the individual deviation μ_{ijt} nonlinearly (through the share integral)

Typical data: Three types of variables

- 1. Market shares (or quantities)
 - Aggregations of individual consumer choices
 - Convert quantities to shares: $s_{jt} = q_{jt} / I_t$ where I_t is market size
 - Key assumption: consumers in the same market face same prices, characteristics
- 2. Prices and product characteristics
 - p_{jt} : prices of each product j in market t
 - x_{jt} : observed characteristics (e.g., horsepower, fuel efficiency)

Typical data: Consumer demographics

- 3. Information on consumer demographics D_i
- Three common data structures:
 - **Micro-data:** Observe D_i for each consumer
 - e.g., survey data on car purchases with buyer demographics
 - **Aggregate data:** Distribution of demographics $F_t(D)$
 - e.g., income distribution from Census/CPS by city
 - Draw simulated consumers from this distribution
 - **Summary statistics:** Average demographics of buyers
 - e.g., average age of consumers who purchase product j

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(A version of) the estimation algorithm in the background

- We have the model—but how do we actually estimate it?
- **Step 1: Estimate π and δ via maximum likelihood**
 - Maximum likelihood finds the π and δ that best predicts consumer choices
 - What variation in the data identifies π ? What variation identifies δ ?

(A version of) the estimation algorithm in the background

- We have the model—but how do we actually estimate it?
- **Step 1: Estimate π and δ via maximum likelihood**
 - Maximum likelihood finds the π and δ that best predicts consumer choices
 - What variation in the data identifies π ? What variation identifies δ ?
 - **Answer:** How choices vary with demographics identifies π ; matching the empirical market shares identifies δ
- **Step 2: Recover δ parameters (β and α) via two stage least squares (2SLS)**
 - Use estimated mean utilities: $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt}$
 - Instruments handle price endogeneity
- **Key:** pyblp handles all of this for you! (using a slightly different algorithm, but similar idea)

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Nevo (2001): The cereal industry

- “Measuring Market Power in the Ready-to-Eat Cereal Industry”
- Why study cereal?
 - Concentrated market (few major firms)
 - High profit margins
 - High advertising-to-sales ratios
 - Aggressive new product introduction
- Classic example of differentiated products industry
- **Question:** Is there collusive pricing?



Nevo (2001): Data and approach

- Utility model is exactly what we have seen:

$$u_{ijt} = x_{jt}\beta_i + \alpha_i p_{jt} + \zeta_{jt} + \epsilon_{ijt}$$

- **Data:**
- Use top 25 cereal brands
- Scanner data → get market shares, prices, etc.
 - Aggregate to MSA-quarter level = 1124 markets.
- Advertising data
- Cereal box characteristics (nutritional information); subjective characteristics ('mushy')
- Demographics from CPS

Nevo (2001): Demand estimates

TABLE VI
RESULTS FROM THE FULL MODEL^a

Variable	Means (β 's)	Standard Deviations (σ 's)	Interactions with Demographic Variables:			
			Income	Income Sq	Age	Child
Price	-27.198 (5.248)	2.453 (2.978)	315.894 (110.385)	-18.200 (5.914)	—	7.634 (2.238)
Advertising	0.020 (0.005)	—	—	—	—	—
Constant	-3.592 ^b (0.138)	0.330 (0.609)	5.482 (1.504)	—	0.204 (0.341)	—
Cal from Fat	1.146 ^b (0.128)	1.624 (2.809)	—	—	—	—
Sugar	5.742 ^b (0.581)	1.661 (5.866)	-24.931 (9.167)	—	5.105 (3.418)	—
Mushy	-0.565 ^b (0.052)	0.244 (0.623)	1.265 (0.737)	—	0.809 (0.385)	—
Fiber	1.627 ^b (0.263)	0.195 (3.541)	—	—	—	-0.110 (0.0513)
All-family	0.781 ^b (0.075)	0.1330 (1.365)	—	—	—	
Kids	1.021 ^b (0.168)	2.031 (0.448)	—	—	—	
Adults	1.972 ^b (0.186)	0.247 (1.636)	—	—	—	
GMM Objective (degrees of freedom)			5.05 (8)			

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What is pyblp?

- Python package for demand estimation
- Conlon & Gortmaker (2020), “Best Practices for Differentiated Products Demand Estimation with PyBLP”
- Handles:
 - Basic logit and logit with demographics
 - Instrumental variables
 - Standard errors
 - Post-estimation (elasticities, markups, etc.)

Installing and importing packages

- **Install once** (in terminal):

```
pip install pyblp numpy pandas
```

- **Import at top of every script:**

```
import pyblp          # demand estimation  
import numpy as np    # numerical arrays (np.array, np.zeros, etc.)  
import pandas as pd   # data frames (like Excel spreadsheets)
```

- np and pd are standard abbreviations

pyblp workflow

1. **Set up data:** products, markets, shares, characteristics
2. **Define formulation:** which variables, which IVs
3. **Create problem:** combine data and formulation
4. **Solve:** estimate the model
5. **Extract results:** coefficients, standard errors, elasticities

Let's walk through each step with car data

Step 1: Load and inspect data

- Load the BLP automobile data
- Key columns: market_ids, shares, prices, hpwt, air, mpd, space, demand_instruments0, ...

```
import pyblp
import pandas as pd

product_data = pd.read_csv(pyblp.data.BLP_PRODUCTS_LOCATION)
```

What data do we need?

- **Product data** (for each product j in market t):
 - Market share s_{jt}
 - Price p_{jt}
 - Characteristics x_{jt} (size, HP, fuel efficiency, etc.)
 - Instruments z_{jt} (BLP IVs, cost shifters)
- **Demographic data** (optional, for interactions):
 - Census data on income, family size, age by market
 - Or micro-level survey data

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The BLP automobile data

market_ids	shares	prices	hpwt	air	mpd	demand_instruments0	demand_instruments1
1971	0.0012	4.94	0.52	0	1.89	2.31	0.48
1971	0.0011	5.52	0.49	0	1.94	2.28	0.51
1971	0.0006	7.11	0.47	0	1.72	2.15	0.53
1971	0.0038	4.30	0.43	0	2.45	1.98	0.57
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

- 2,217 products across 20 years (1971–1990)
- BLP instruments: sums of rivals' characteristics

Model 1: Basic logit specification

- **Utility:**

$$u_{ijt} = \beta_1 \cdot \text{hpwt}_{jt} + \beta_2 \cdot \text{air}_{jt} + \beta_3 \cdot \text{mpd}_{jt} + \beta_4 \cdot \text{space}_{jt} + \alpha \cdot p_{jt} + \zeta_{jt} + \varepsilon_{ijt}$$

- **Parameters to estimate:**

- $\beta_1, \beta_2, \beta_3, \beta_4$: tastes for characteristics
- α : price sensitivity (expect $\alpha < 0$)

- **Note:** All consumers have the same preferences (no heterogeneity)

Estimating Model 1

- Define the formulation and solve

```
# Define formulation: characteristics + price (no constant)
formulation = pyblp.Formulation('0 + hpwt + air + mpd + space + prices
    ')

# Create problem (pyblp detects instruments automatically)
problem = pyblp.Problem(formulation, product_data)

# Estimate using 2SLS
results = problem.solve()
```


Working with results

- **Print summary:** shows coefficients, std errors, diagnostics

```
print(results)  # displays formatted table
```

- **Access specific attributes:**

```
results.beta          # linear coefficients (numpy array)  
results.beta_se       # standard errors  
results.xi            # estimated unobservables (xi_jt)
```

Working with results: post-estimation

- **Compute elasticities:**

```
elasticities = results.compute_elasticities()  
# Returns matrix: elasticities[j,k] = % change in q_j from 1% price  
change in k  
  
own_price_elasticities = np.diag(elasticities) # extract diagonal
```

- Own-price elasticity: η_{jj} on diagonal
- Cross-price elasticity: η_{jk} off diagonal
- `np.diag()` extracts the diagonal of a matrix

Model 1: Results

Variable	Coefficient	Std. Error
HP/Weight	-1.251	(0.623)
Air conditioning	1.773	(0.194)
Miles per dollar	-1.245	(0.047)
Space	-1.643	(0.097)
Price (α)	-0.225	(0.017)

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Agent data: Consumer demographics

- pyblp uses “agent data” for demographic draws:

market_ids	weights	income
1971	0.0005	1.10
1971	0.0007	0.45
1971	0.0005	1.27
1971	0.0007	0.23
⋮	⋮	⋮

- **weights**: Integration weight for each draw (sum to 1 per market)
- **income**: Income in \$100,000s (e.g., 1.10 = \$110k)
- pyblp integrates over these draws to compute market shares

Model 2: Adding demographic interactions

- **Utility with income interactions:**

$$u_{ijt} = \delta_{jt} + \underbrace{(\pi_{\text{inc}} \cdot \text{income}_i) \cdot p_{jt}}_{\mu_{ijt}} + \varepsilon_{ijt}$$

- Where $\delta_{jt} = \beta_1 \cdot \text{hpwt}_{jt} + \dots + \alpha_0 \cdot p_{jt} + \xi_{jt}$
- **New parameter:** π_{inc} (income $\times$ price)
 - If $\pi_{\text{inc}} > 0$: high-income less price-sensitive
- **Linear:** $\beta_1, \dots, \alpha_0$ (in δ_{jt}) **Nonlinear:** π_{inc} (in μ_{ijt})

Estimating Model 2

```
# Scale income to $100,000s for numerical stability
agent_data['income'] = agent_data['income'] / 100

X1 = pyblp.Formulation('0 + hpwt + air + mpd + space + prices')
X2 = pyblp.Formulation('0 + prices') # interacts with demographics
agent_formulation = pyblp.Formulation('0 + income')

problem = pyblp.Problem(
    (X1, X2), product_data, agent_formulation, agent_data
)
results = problem.solve(sigma=0, pi=np.array([[0.5]]))
```

Understanding the solve command

- `results = problem.solve(sigma=0, pi=np.array([[0.5]]))`
- **numpy** (`np`): Python library for numerical arrays
 - `np.array([[0.5]])` creates a matrix with value 0.5
- **sigma**: `sigma=0` means no random coefficients
 - We use observed demographics only
- **pi**: Demographic interaction parameters (π)
 - Starting values for the optimizer (initial guess)
 - `pyblp` finds the best values via GMM
- **results**: Object containing all estimation output

Model 2: Results

Parameter	Estimate	Interpretation
<i>Linear parameters (in δ_{jt}):</i>		
α_0 (price)	-0.728	Base price sensitivity
<i>Demographic interaction (in μ_{ijt}):</i>		
π_{inc} (income $\times$ price)	+0.014	High-income less sensitive

- Effective price coef: $\alpha_0 + \pi_{\text{inc}} \cdot \text{income}_i$
- Income = \$50k: $-0.728 + 0.014 \times 0.5 = -0.721$
- Income = \$200k: $-0.728 + 0.014 \times 2.0 = -0.700$ (less price-sensitive)

Elasticity matrix example

	Prod 1	Prod 2	Prod 3	Prod 4	Prod 5
Prod 1	-3.5	0.02	0.03	0.01	0.02
Prod 2	0.02	-4.1	0.03	0.02	0.03
Prod 3	0.01	0.02	-2.8	0.01	0.02
Prod 4	0.02	0.03	0.02	-3.9	0.02
Prod 5	0.01	0.02	0.01	0.01	-3.2

- **Diagonal (red):** Own-price elasticities (negative, $|\eta| > 1$)
- **Off-diagonal:** Cross-price elasticities (positive for substitutes)
- Entry (j, k) : % change in s_j from 1% increase in p_k

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Worked example: Interpret this output

- pyblp gives you this output:

Variable	Estimate	Std. Error
Constant	3.5	0.8
HP/weight	1.2	0.4
Air cond.	0.9	0.3
Price	0.05	0.02

- **Question:** Something is wrong. What is it?

Take 1 minute to identify the problem.

Worked example: Interpret this output (solution)

- **Problem:** Price coefficient is positive (+0.05)!
 - Higher prices $\rightarrow$ higher utility. Economically nonsensical.
- **Why might this happen?**
 1. Weak instruments: IVs don't predict prices well
 2. Wrong signs: Maybe you forgot a negative sign somewhere
 3. Endogeneity still present: IVs are correlated with ξ
- **Fix:** Check first-stage F-stat, try different IVs, verify data

Common pyblp errors and how to fix them

- **“Singular matrix” error:**

Common pyblp errors and how to fix them

- **“Singular matrix” error:**
 - Likely: collinear variables in formulation
 - Fix: Remove redundant variables
- **Price coefficient positive or near zero:**

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Common pyblp errors and how to fix them

- **“Singular matrix” error:**
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 - Fix: Remove redundant variables
- **Price coefficient positive or near zero:**
 - Likely: weak instruments
 - Fix: Check first-stage F-stat; try different IVs
- **Huge standard errors:**
 - Likely: weak identification or too few observations
 - Fix: Aggregate to fewer markets; add more IVs

How to check your results make sense

1. **Price coefficient negative?** (Required!)
2. **Elasticities reasonable?**
 - Own-price: typically -2 to -10 for durables
 - Cross-price: positive but small
3. **Markups reasonable?**
 - Cars: typically 15-30%
 - Cereal: typically 30-50%
4. **First-stage F-stat > 10 ?** (IV relevance)
5. **Coefficients stable across specifications?**

This prepares you for HW1

- HW1 asks you to:
 1. Estimate basic logit (like Model 1)
 2. Estimate with demographics (like Model 2)
 3. Compute elasticities
 4. Interpret your results
- Key differences in HW1:
 - Uses cereal data (not cars)
 - Simpler: just sugar + price (not 5 characteristics)
 - Same structure: Model 1 $\rightarrow$ Model 2

Key Points

1. **pyblp**: Python package for demand estimation (Conlon & Gortmaker 2020)
2. **Workflow**: Data $\rightarrow$ Formulation $\rightarrow$ Problem $\rightarrow$ Solve $\rightarrow$ Interpret
3. **Key outputs**: Coefficients (check signs), standard errors, elasticities, markups
4. Price coefficient should be **negative**; OLS biases it toward zero
5. Demographics: interact with product characteristics to capture preference heterogeneity
6. This is the foundation for HW1